

2024



# Improving Sustainability and Price Performance Using AWS Graviton–Based Instances with Pinterest

Learn how Pinterest improved infrastructure cost efficiency and environmental sustainability using AWS Graviton–based instances.

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**38% savings**

on compute resources for an important workload

**47% cost savings**

for an important workload

**25%**

of infrastructure migrated to Graviton instances with s



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# Overview

[Pinterest](#) is a visual discovery platform that people worldwide use to find products that are personalized to their tastes. The platform is available in over 40 languages, and they have more than half a billion monthly active users who visit it to explore and experience billions of ideas and inspiring content. Pinterest is mindful that the inspiration that it hopes to bring to the world depends on individual wellbeing, so the company is committed to doing its part to address climate change risks. Pinterest is committed to taking action through science-based emissions-reduction targets, preserving natural resources, and collaborative partnerships to expand its positive impacts.

As a born-in-the-cloud company, Pinterest has always run on Amazon Web Services (AWS). In 2021, Pinterest launched a program to migrate workloads to [AWS Graviton-based Amazon EC2 instance](#), which deliver excellent price performance for a broad range of workloads running on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. As Pinterest scaled up its use of AWS Graviton-based Amazon EC2 instances through 2024, the company improved its infrastructure cost efficiency while maintaining its strong commitment to energy efficiency, emissions reduction, and sustainability.



## Opportunity | Promoting Sustainability and Increasing ROI by Migrating to AWS Graviton-Based Instances

Pinterest's mission is to bring everyone the inspiration to create a life they love. This mission requires building handy applications that use artificial intelligence to retrieve information from a very large infrastructure. While the company is always looking for ways to improve its overall return on investment (ROI) from this infrastructure, it was also simultaneously achieving its sustainability goals, and it saw an opportunity to do so by adopting AWS Graviton-based instances.



“We look for scenarios where we can improve sustainability and ROI at the same time,” says David Chaiken, the chief architect of Pinterest. “We deeply appreciate the move by AWS to renewable energy, which helps us be more efficient and reduce the total energy that we’re using.”

Pinterest had already been using Amazon EC2 for almost 10 years when AWS announced AWS Graviton–based instances in 2019. The company evaluated how AWS Graviton–based instances could benefit its workloads and did ROI studies, which led to investment in a migration program in 2021. The first step in the process was to build a small, cross-functional team of domain experts to prepare its code base, and build and deploy systems for the change. The next step was to gain experience with product deployments and benchmark four public workloads in 2022. The migration ramped up with an engineering-wide program throughout 2023–2024, and by June 2024, the team migrated more than 25 percent of Pinterest’s compute infrastructure.

Working closely alongside the AWS team for many years, Pinterest has been able to accelerate the migration to AWS Graviton–based instances. Throughout the process, AWS experts provided guidance on AWS Graviton, including ROI and sustainability analysis through workload deployments. “Working alongside AWS, we aligned our sustainability goals with our business goals,” says Sierra Hutto, senior technical program manager at Pinterest, who drove the migration initiative.

## Solution | Improving Price Performance and Producing 62 Percent Lower Carbon Emissions Using AWS Graviton–Based Instances

As part of the migration, Pinterest identified an important workload and worked alongside the AWS team to benchmark the positive impact of AWS Graviton–based instances on improving workload efficiency. The workload consists of thousands of instances that serve Pinterest’s API, which is central to every aspect of Pinterest’s product and business. These instances respond to billions of API requests per day. For this workload, Pinterest found that the migration to AWS Graviton–based instances resulted in a 38 percent reduction in compute resources, 62 percent lower carbon emissions per API request, and 47 percent reduction in workload costs.

The results for this specific workload might not apply across all the migrated workloads, but Pinterest has realized a meaningful ROI and sustainability impact from the migration to AWS Graviton–based instances. “We had to choose between business outcomes and sustainability goals, because using AWS Graviton, we were able to improve both.”



cost efficiency while emitting less carbon,” says Chris Bunting, engineering manager at Pinterest, whose team runs Pinterest’s client APIs.

The work on migrating to AWS Graviton-based instances is one of Pinterest’s many cost-efficiency programs that align business with sustainability goals. This effort reduces overall resource consumption and is complemented by the transition to renewable energy. Using renewable energy is critical for Pinterest’s commitment to its emissions-reduction targets. “We’ve grown Pinterest’s monthly active users to record highs, while continuing to make progress toward our sustainability goals,” says Mia Ketterling, senior corporate affairs lead for environmental sustainability at Pinterest. The company’s [2024 ESG Impact Report](#) provides more information on its global work on sustainability.

## Outcome | Continuing to Achieve Further Sustainability and Performance Improvements

Pinterest’s migration to AWS Graviton-based instances was part of a comprehensive program to modernize and rightsize the company’s compute infrastructure. A central component of the program’s success was including AWS Graviton in the set of approved instances—empowering the engineering team to continue to innovate while achieving bottom-line goals. “Our investment in AWS Graviton-based instances has paid off significantly,” says Madhuri Racherla, vice president of infrastructure engineering at Pinterest. Pinterest is now looking forward to further efficiency gains as additional workloads are deployed on AWS Graviton-based instances.

“When it comes to climate change, a lot of relatively small projects add up to a real difference,” says Chaiken. “Using AWS Graviton-based instances is one of those, and we’re proud of the work that we’ve done alongside the AWS team.”

## About Pinterest

Pinterest is a visual search and discovery platform where people find inspiration, curate ideas, and shop products—all in a positive place online.

## AWS Services Used

### Amazon Elastic Compute Cloud (Amazon EC2)

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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## AWS Graviton Processor

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

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